Lesson 9: Fuses and Earthing

Teacher: Dan

Format: Theory & 1 Mandatory Practical

Topic: Domestic Electricity and Electrical Safety

Gemini Notebook

GCSE PHYSICS P3.2 · DOMESTIC USES & ELECTRICAL SAFETY

Starter Activity: Prior Knowledge Retrieval Grid

5-10 MINS · 6 KNOWLEDGE CHECKS

QUESTION 1 · POWER CALCULATIONS

P = I × V

State the equation linking power (P), current (I), and potential difference (V). Calculate the power of a toaster running at 230 V that draws a current of 4 A.

ANSWER P = I × V = 4 A × 230 V = 920 W (requires a 5 A or 13 A fuse).

QUESTION 2 · POWER & RESISTANCE

$P = I^2 \times R$

State the equation linking power (P), current (I), and resistance (R). By what factor does heating power increase if the current flowing through a resistor triples?

ANSWER $P = I^2 \times R$. Power is proportional to current squared (I^2), so tripling current increases power by 9× ($3^2 = 9$).

QUESTION 3 · CURRENT FLOW & CHARGE

Q = I × t · Charge rate

Define electric current and state the equation linking charge (Q), current (I), and time (t). How does the direction of electron flow compare to conventional current?

ANSWER Current is the rate of flow of electric charge (Q = I × t). Electrons flow from negative to positive; conventional current flows positive to negative.

QUESTION 4 · 3-PIN PLUG PINS

Brown · Blue · Green/Yellow

Identify the 3 pins in a UK plug (Live, Neutral, Earth) by their positions and wire colours. Which pin connects directly to the fuse, and why?

ANSWER Live = Brown (bottom right, connects to fuse); Neutral = Blue (bottom left); Earth = Green/Yellow (top center). Fuse isolates live 230 V supply.

QUESTION 5 · PLUG PIN SAFETY

Safety shutters · Brass hardness

Why is the earth pin made longer than the live and neutral pins, and why are all plug pins manufactured from solid brass?

ANSWER Longer earth pin opens socket safety shutters and connects earth first for protection. Brass is a rigid, corrosion-resistant, high conductor.

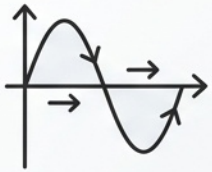
QUESTION 6 · WHY TRANSFORMERS ARE USED

High V → Low I → Low I^2R loss

Explain why step-up and step-down transformers are used in the National Grid transmission system.

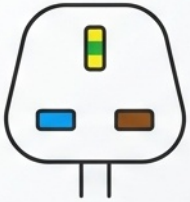
ANSWER Step-up transformers increase voltage (up to 400 kV) to reduce current, drastically cutting thermal energy losses ($P = I^2R$) in cables. Step-down reduces to 230 V for safety.

Learning Objectives



Context

- **Understand** UK mains supply.
- **Compare** alternating and direct current.



Interface

- **Identify** 3-pin plug wiring.
- **Calculate** correct fuse ratings.





Protection

- **Explain** how earthing prevents shocks.
- **Synthesise** the sequence of fault protection.

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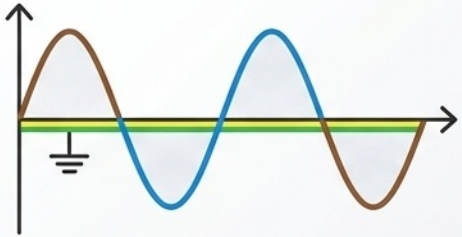
The UK Mains Supply



Direct Current (DC): Voltage remains constant.

Source: Batteries and solar cells.

Direction: Charge flows in one direction only.



Alternating Current (AC): Voltage continuously changes direction.

UK Standard: $230\ \mathrm{V}$ potential difference.

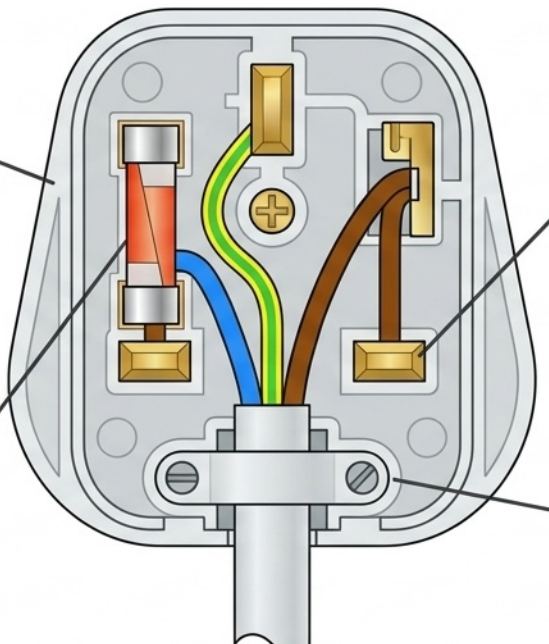
Frequency: $50\ \mathrm{Hz}$ (changes direction 50 times a second).

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Anatomy of the 3-Pin Plug

Insulating Casing: Stiff plastic or rubber prevents user contact with live components.

Fuse Location: Sits in series with the live pin before the appliance switch.



Brass Pins: Excellent electrical conductor, highly hard-wearing.

Cable Grip: Secures the outer sheath, preventing wires from pulling loose.

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Inside the Cable

Live Wire

- **Colour:** Brown (Bottom Right).
- **Voltage:** 230 V alternating potential difference.
- **Function:** Carries power into the appliance.
- **Danger:** Most dangerous wire.

Neutral Wire

- **Colour:** Blue (Bottom Left).
- **Voltage:** Held at or close to 0 V.
- **Function:** Completes the electrical circuit.

Earth Wire

- **Colour:** Green and Yellow stripes (Top Centre).
- **Voltage:** 0 V (connected to ground).
- **Function:** Safety wire connected to metal casing.

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The Fuse





- **Structure:** A thin metal wire inside a ceramic or glass tube.
- **Property:** Wire has a deliberately low melting point.
- **Position:** Always connected in **series** with the Live wire.
- **Standard UK Ratings:** 3 A, 5 A, and 13 A.

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How a Fuse Protects

1: Surge

Current exceeds the specific rating of the fuse.

2: Heating

Electric current causes a heating effect.

$$P = I^2R$$

3: Melt

The thin wire rapidly heats up, melts, and blows.

4: Isolate

The circuit breaks, disconnecting the 230 V live supply.

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Choosing the Right Fuse

$$I = P / V$$

Step 1: Identify

- Find appliance Power (P) in Watts.
- UK Mains Voltage (V) is always 230 V.

Step 2: Calculate

- Use equation: I = P / V
- Example 1: Kettle (2300 W) / 230 V = **10 A**
- Example 2: Toaster (690 W) / 230 V = **3 A**

Step 3: Select

- Choose the standard fuse immediately **above** the normal operating current.
- Kettle: Needs **13 A fuse**.
- Toaster: Needs **5 A fuse**.

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Knowledge Checkpoint

Card 1: Wiring

Which colour wire is connected to the bottom right pin, and what is its voltage?

(Wait for class response: Brown, Live, 230 V)

Card 2: Calculation

A microwave operates at 1150 W.

Calculate the current ($I = P/V$).

Which fuse should you fit: 3 A, 5 A, or 13 A?

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The Hazard: Electric Shock

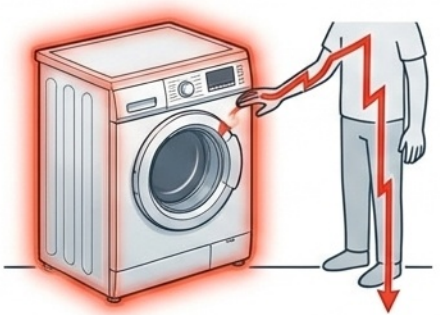


- **The Fault:** A live wire comes loose and touches the metal casing.
- **The Danger:** The entire metal casing becomes live at 230 V.
- **The Shock:** Touching the casing provides a path to ground through the human body.
- **The Result:** A severe or fatal electric shock.

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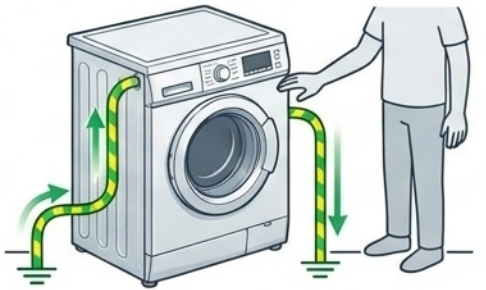
The Hero: Earthing

Without Earth



- Current travels through the path of least resistance.
- Without an Earth wire, that path is the human body.

With Earth



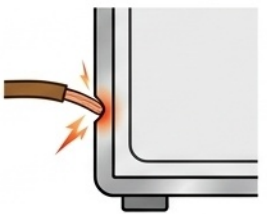
- Provides an extremely low resistance path.
- Current bypasses the human body entirely.

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The Sequence of Protection

1. The Fault

Live wire touches the earthed metal casing.



1

2. The Surge

Current surges straight down the low-resistance Earth wire to ground.



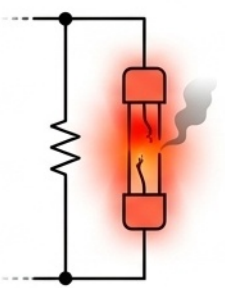
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3. The Break

Massive current

$$I = V/R$$

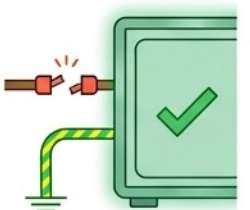
instantly exceeds the **fuse** rating, melting the wire.



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4. The Isolation

Appliance is completely disconnected from the live supply. Safe to touch.



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Double Insulation



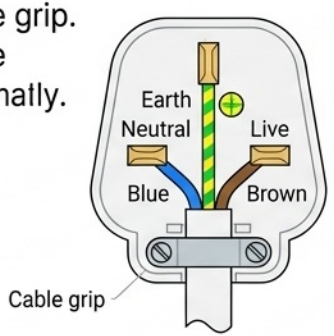
- **The Exception:** Appliances with plastic outer casings (e.g., hair dryers, vacuums).
- **No Metal:** Zero exposed metal parts to become live.
- **No Earth Needed:** Casing cannot shock the user; Earth wire is entirely omitted.
- **The Symbol:** Marked with the double square symbol.
- **Wiring:** Uses only a 2-core cable (Live and Neutral).

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Practical: Fuses & Earthing Workflow

Step 1: Wire the Plug

- Examine and correctly wire a 3-pin plug.
- Check wire colours, pin connections, and secure cable grip. cable griematly.



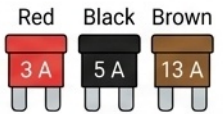
Step 2: Calculate Ratings

- Read power labels on provided appliances (lamp, heater).
- Calculate current and select correct fuse (3 A, 5 A, or 13 A).

LAMP
60 W, 230 V

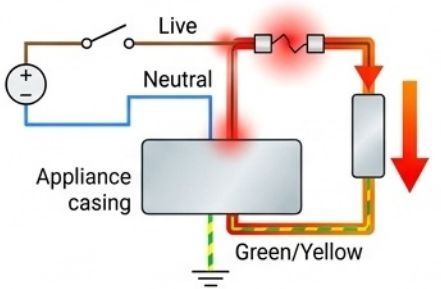
HEATER
2000 W, 230 V

Current $I = \frac{P}{V}$ →



Step 3: Fault Demo

- Observe the teacher's model fault circuit.
- Measure the current surge and watch the fuse blow safely.



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Summary Cheat Sheet

Core Stats

- **UK Mains**: 230 V, 50 Hz (Alternating Current).
- **Fuses**: 3 A, 5 A, 13 A.

The Wires

- **Live (Brown)**: 230 V (**Danger**).
- **Neutral (Blue)**: ~0 V (Completes circuit).
- **Earth (Green/Yellow)**: 0 V (Safety path to ground).

The Protection

- Calculate fuse using:

$$I = P / V.$$

- Earth wire gives low resistance path

→ Surge blows fuse

→ Appliance isolated.

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